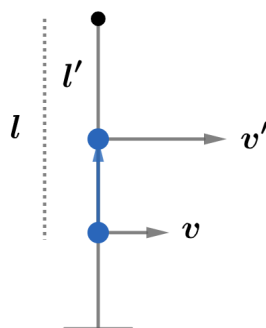


## 2020B F=ma Exam: Problem 21

Kevin S. Huang



Since there are no external torques about the pivot, angular momentum is conserved. Thus,

$$L = mvl = mv'l'$$

$$vl = v'l'$$

so the answer is .